

Portfolio_Lashkarbek_Nuriman

Synopsis: Portfolio Analysis – Toyota Motor Corporation vs Volkswagen AG

This portfolio presents a comparative financial analysis of Toyota Motor Corporation (TM) and Volkswagen AG (VOW3.DE) for the period from March 2025 to March 2026. The objective of the analysis is to evaluate the risk and return characteristics of both companies and compare their performance using statistical and financial metrics.

The dataset includes weekly closing prices and calculated weekly returns for both assets. Based on these data, several statistical methods were applied in order to analyze the behavior and volatility of the two stocks.

First, descriptive statistics were calculated for the weekly returns of both companies. These statistics include the maximum and minimum values, range, quartiles, interquartile range (IQR), mean return, mean absolute deviation, sample variance, standard deviation, and coefficient of variation. In addition, covariance and correlation between the returns of Toyota and Volkswagen were calculated to understand the relationship between the two assets.

Second, the Sharpe Ratio was used to evaluate the risk-adjusted performance of the investments. The Sharpe Ratio compares the excess return of an asset relative to the risk-free rate per unit of risk. In this analysis, the U.S. Treasury Bill rate was used as the risk-free rate and converted to a weekly equivalent in order to match the frequency of the return data.

Furthermore, an F-test for equality of variances was performed to determine whether the volatility of the two stocks is statistically different. This hypothesis test compares the variances of the weekly return distributions and evaluates the results at the 5% significance level.

Finally, regression analysis was conducted using the S&P 500 index as a benchmark for the overall market. The regression compares the weekly returns of the selected asset with the weekly returns of the S&P 500 in order to analyze the relationship between the stock and the market and evaluate market sensitivity.

Overall, this portfolio combines descriptive statistics, risk-adjusted performance metrics, hypothesis testing, and regression analysis to provide a comprehensive comparison of Toyota Motor Corporation and Volkswagen AG.
